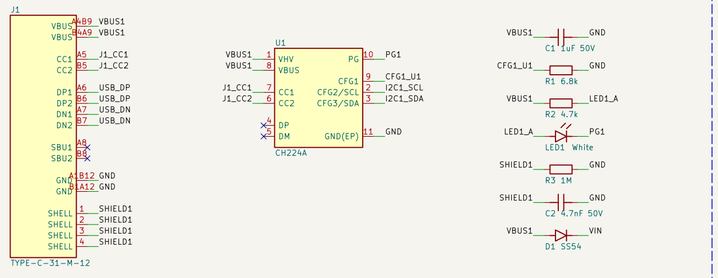
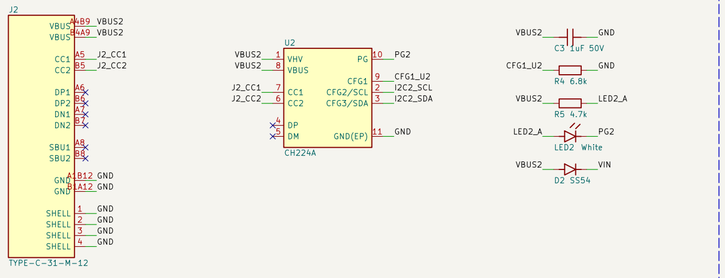
(1) USB-C J1: data + power / U1: PD sink, requests 9V

CH224A: Rset 6.8k on CFG1 = 9V (datasheet table 5-1). VHV and VBUS pins tied to the connector VBUS, 1uF on VHV.
CC1/CC2 go straight to U1 (Rd is inside U1); DP/DM of U1 are not used (data goes to J4 only).
CFG2/CFG3 = I2C (0x22/0x23, address not selectable) -> own bus per chip. PG is open drain, low when a PD contract is made.
D1 blocks back-feed from VIN to the PC. Shield: 1M || 4.7nF to GND.



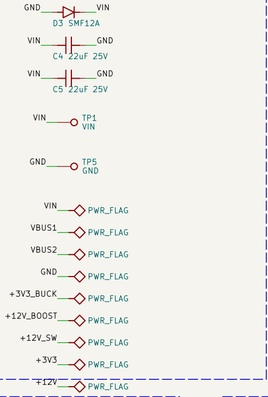
(2) USB-C J2: power only (charger) / U2: PD sink, requests 9V

Same circuit as (1). D+/D- of J2 are not connected. Shield of J2 goes straight to GND.
D2 blocks back-feed from VIN to the charger port; D1/D2 together pick the higher input.



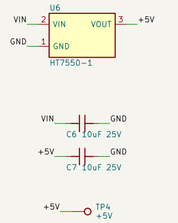
VIN bus (4.2 .. 9V)

TVS SMF12A (12V standoff, 19.9V clamp).
Bulk capacitors sit behind D1/D2
(connector side has only 1uF).



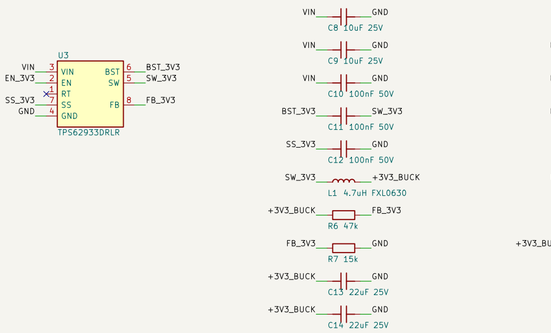
U6: 5V LDO (MCU + J4 VBUS)

HT7550-1: 1=GND 2=VIN(tab) 3=VOUT. 10uF in / 10uF out
(datasheet application circuit).
At VIN=5V the output follows VIN minus dropout
(about 4.2 .. 4.9V).



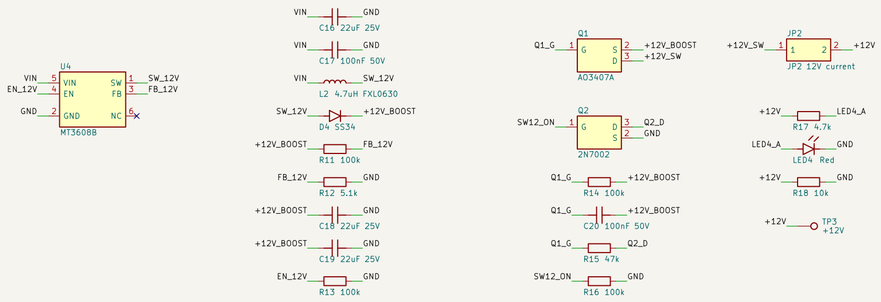
U3: 3.3V / 3A buck (TPS62933) + EN from J1 VBUS + JP1 (current measurement)

Vout = 0.8V x (1 + 47k/15k) = 3.307V. RT open = 500kHz. L = 4.7uH (datasheet table 10-2). SS 100nF -> tSS = 100n x 0.8 / 5.5u = 14.5ms.
EN = VBUS1 x 68k/(100k+68k) = 0.405 x VBUS1: turns on at VBUS1 > 3.0V typ (3.16V max). EN = 2.0V @5V, 3.6V @9V, 5.5V @13.6V.
D5 clamps EN to +5V + VF (< 6V abs max) if VBUS1 ever goes higher. MCU (PC0, open drain) can pull EN low.
Charger on J2 alone -> VBUS1 = 0 -> EN low -> 3.3V stays off.



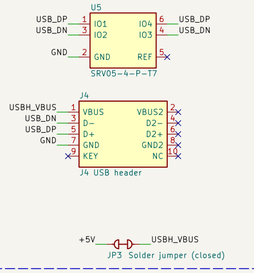
U4: 12V / 0.5A boost (MT3608B) + Q1/Q2 output switch + JP2 (current measurement)

Vout = 0.6V x (1 + 100k/5.1k) = 12.36V. L = 4.7uH. D4 = SS34 (datasheet application). EN pulled down 100k -> off with a blank MCU.
A boost passes VIN to its output when stopped, so Q1 (P-ch, Vgs max +-20V) cuts the output. Q2 pulls Q1 gate down through 47k:
Vgs = -12.36 x 100k/147k = -8.4V. 100nF gate-source = soft turn-on (about 1.6ms). Q2 gate pulled down 100k -> off with a blank MCU.
R18 bleeds the +12V line to 0V when off.



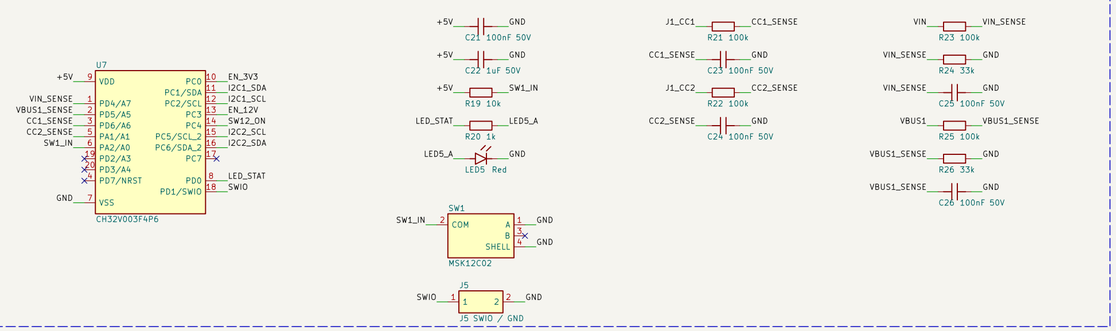
USB 2.0 data: J1 -> U5 (ESD) -> J4

SRV05-4: 1=IO1 2=GND 3=IO2 4=IO3 5=REF 6=IO4.
D+ passes pins 1-6, D- passes pins 3-4.
REF is left open: its TVS breaks down at 6V.
VBUS1 can be 9V.
J4 = motherboard USB 2.0 header pinout:
1 VBUS, 3 D-, 5 D+, 7 GND, 9 key (pull the pin).
JP3 (solder jumper, closed) feeds +5V to J4 pin 1.



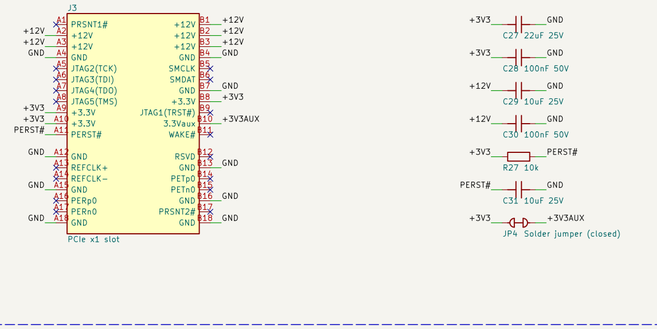
U7: CH32V003F4P6 — reads CC / VIN / VBUS1, switches 3.3V and 12V

VDD = +5V (4.2 .. 5V). ADC reference = VDD, so firmware corrects with the internal Vref channel.
CC1/CC2: 100k series + 100nF (the 100k keeps the load off the PD line: fault current into the pin < 0.2mA).
VIN / VBUS1 sense = x 33k/(100k+33k) = 0.248: 5V -> 1.24V, 9V -> 2.23V, 20V (TVS clamp) -> 4.96V (below VDD = 5V whenever VIN > 5.5V).
PC0 = EN_3V3 (drive low or leave Hi-Z, never push-pull high). PC3 = EN_12V. PC4 = SW12_ON (both have 100k pull-downs).
I2C1 (PC1/PC2) -> U1. Remap I2C1 to PC5/PC6 (AFIO I2C1_RM = 10b) -> U2. No external pull-ups: CH224A has them inside.
J5: 1 = SWIO, 2 = GND (WCH-LinkE). SW1: closed to GND = manual 12V override.



J3: PCI Express x1 slot — power pins only

Pin names follow the PCIe CEM pinout (side B = component side of the card, side A = solder side).
+12V: A2 A3 B1 B2 B3. +3.3V: A9 A10 B8. 3.3Vaux: B10 via JP4 (solder jumper, closed).
GND: A4 A12 A15 A18 B4 B7 B13 B16 B18. PERST# (A11): 10k to +3.3V, 10uF to GND -> rises ~93ms after 3.3V.
PRSNT#, WAKE#, JTAG, SMBus, REFCLK and the lane pins are left open.



Mechanical

M3 mounting holes (not plated,
no net).



Generated by scripts/gen_schematic.py — edit the script, not this file
USB-C x2 -> 3.3V / 12V for USB-data PCIe tuner cards (PX-MLT5PE etc.)
danything

Sheet: /
File: pcie-tuner-board.kicad_sch

Title: PCIe tuner external power board

Size: A2	Date: 2026-11-11	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 1/1